FINAL TECHNICAL REVIEW

NAME:- SREEJA SANYAL
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Introduction and Project Overview

- This project focuses on lithium-ion battery charging data.
- It uses machine learning to estimate the battery State of Charge (SOC).
- It also predicts the Remaining Charging Time.
- The project aims to provide reliable predictions using battery parameters.

Final Methodology and Validation

- Dataset preprocessing and feature selection were completed.
- XGBoost Regressor was selected for SOC estimation.
- Random Forest Regressor was selected for Remaining Charging Time prediction.
- Model performance was evaluated using MAE, RMSE and R^2.
- Final model selection was based on validation performance.

Workflow and Error Analysis

- **Input-to-Output Workflow:**

Dataset → Preprocessing → Feature Selection → Model Training → Evaluation → Prediction

- **Error Analysis:**

1. Prediction errors were reviewed using model evaluation outputs.

1. Performance may vary under different battery operating conditions.

1. Model results and observations were recorded for reporting.

Thank You